

Peiqi Wang

wpq@mit.edu • 32 Vassar Street 32-D472, Cambridge, MA 02139 • people.csail.mit.edu/wpq

Education

Massachusetts Institute of Technology	Cambridge, MA
Ph.D. in Electrical Engineering and Computer Science	Sep 2019 – May 2025
S.M. in Electrical Engineering and Computer Science	Sep 2019 – Feb 2022
University of Toronto	Toronto, ON
B.S. in Computer Science, with High Distinction. GPA: 3.95/4.0	Sep 2014 – May 2019

Industry Experience

Meta	Jun 2024 – Dec 2024
Research Scientist Intern	Menlo Park, CA
Investigated the optimal allocation of inference compute across key scaling factors in video vision language models. Conducted large-scale training sweeps and parametric modeling of task performance to identify the inference compute-optimal frontier. [4]	
MIT-IBM Watson AI Lab	Jun 2023 – Aug 2023
Research Intern	Cambridge, MA
Developed a dataset diversity measure strongly correlated with downstream instruction-following performance. Leveraged determinantal point processes to quantify diversity and optimize subset selection for instruction-tuning datasets. [5]	
Deep Genomics	Sep 2017 – Aug 2018
Research Student & Summer Intern	Toronto, ON
Improved GenomeKit, a C++ library for genomic analysis, by extending its functionality. Developed and benchmarked a high-speed tool for approximating energy-based RNA-RNA interaction predictors. Improved a key model for alternative splicing.	

Research Experience

Medical Vision Group, MIT CSAIL	Sep 2019 – May 2025
Research Assistant	Advisor: Polina Golland. Cambridge, MA
- Proposed a novel approach to assess and improve the calibration of humans and computational models that communicate confidence using linguistic expressions of certainty [3]. - Improved vision language pretraining methods to better capture local correspondences [10] and to alleviate the performance drop from sampling false negatives [9]. - Developed a method that injects domain knowledge to construct models that provide clinically feasible supporting evidence. [11]	
Dynamic Graphics Project, University of Toronto	Sep 2018 – May 2019
Research Assistant	Advisor: Kyros Kutulakos. Toronto, ON
Developed structured light reconstruction algorithms for coded-exposure camera.	
Numerical Analysis Group, University of Toronto	May 2018 – Nov 2018
Research Assistant	Advisor: Kenneth Jackson. Toronto, ON
Analyzed and resolved an inconsistency in a two-level importance sampling algorithm for simulating portfolio credit risk under the Gaussian Copula model. Developed and implemented Monte Carlo methods with variance reduction techniques in Julia to improve efficiency.	

Teaching Experience

MIT 6.3700/6.3702 Introduction to Probability	Sep 2022 – Dec 2022
Teaching Assistant	
Designed problem sets and exam questions. Led office hours and tutorials for ~ 80 students.	

Academic Service

Organizer: Boston Medical Imaging Workshop 2024

Conference Reviewer: MedNeurips 22-23, AI4Science 23, MICCAI 23-25, ML4H 23-24, WACV 24-25, MIDL 24-25, NeurIPS 24, ECCV 24, ICLR 25, CVPR 25, ICCV 25, MLHC 25, ACL 25

Journal Reviewer: IEEE Transactions on Medical Imaging (TMI), Medical Image Analysis (MedIA), Machine Learning for Biomedical Imaging (MELBA)

Skills

Programming Languages: Python, C/C++, Julia, R, Racket, MATLAB, JavaScript, $\text{\LaTeX}$.

Libraries and Tools: PyTorch, NumPy, Pandas, Matplotlib, Jax, Linux, Slurm, Git, SQL, MongoDB, React, Node.js, HTML, CSS.

Honors and Awards

Takeda Fellowship (2023), Dean's List (2015-2019), Lawrence And Sharen Ho International Scholarship (2017), The Kathryn Anne Radford Scholarships (2016), The Milne Research Award (2016), Adel S Sedra Undergraduate Scholarship In Mathematics (2015), University of Toronto Scholars Awards (2014).

Publications

1. S. Peng, K. S. Tai, B. He, X. Chen, **P. Wang**, T. Tian, Q. Zhang, S. N. Shukla, L. Zhang, D. H. Chau, and D. Yang. Accelerating LLM-Based Text Embeddings: The Power of Global Tokens for Efficient Token Reduction.
2. Y. Liu, **P. Wang**, S. Diaz, B. Billot, E. A. Turk, E. Grant, and P. Golland. Fetuses Made Simple: Modeling and Tracking of Fetal Shape and Pose. In: *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. 2025.
3. **P. Wang**, B. D. Lam, Y. Liu, A. Asgari-Targhi, R. Panda, W. M. Wells, T. Kapur, and P. Golland. Calibrating Expressions of Certainty. In: *International Conference on Learning Representations (ICLR)*. 2025.
4. **P. Wang**, S. Peng, X. Zhang, H. Yu, Y. Yang, L. Huang, F. Liu, and Q. Wang. Inference Compute-Optimal Video Vision Language Models. In: *Annual Meeting of the Association for Computational Linguistics (ACL)*. 2025.
5. **P. Wang**, Y. Shen, Z. Guo, M. Stallone, Y. Kim, P. Golland, and R. Panda. Diversity Measurement and Subset Selection for Instruction Tuning Datasets. In: *ICLR DATA-FM workshop*. 2025.
6. B. D. Lam, S. Ma, I. Kovalenko, **P. Wang**, O. Jafari, A. Li, and S. Horng. MIMIC-IV-Ext-PE: Using a Large Language Model to Predict Pulmonary Embolism Phenotype in the MIMIC-IV Dataset. In: *Research and Practice in Thrombosis and Hemostasis (RPTH)*. 2024.
7. B. D. Lam, **P. Wang**, S. Ma, O. Jafari, I. Kovalenko, and A. Li. Large language models for chart review: How machine learning can accelerate hematology research. In: *Blood VTH*. 2024.
8. Z. Guo, **P. Wang**, Y. Wang, and S. Yu. Improving Small Language Models in Domain-Specific QA via Generative Data Augmentation. In: *KDD LLM4AI Workshop*. 2023.
9. **P. Wang**, Y. Liu, C.-Y. Ko, W. M. Wells, S. Berkowitz, S. Horng, and P. Golland. Sample-Specific Debiasing for Better Image-Text Models. In: *Machine Learning for Healthcare Conference (MLHC)*. 2023.
10. **P. Wang**, W. M. Wells, S. Berkowitz, S. Horng, and P. Golland. Using Multiple Instance Learning to Build Multimodal Representations. In: *Information Processing in Medical Imaging (IPMI)*. 2023.
11. **P. Wang**, R. Liao, D. Moyer, S. Berkowitz, S. Horng, and P. Golland. Image Classification with Consistent Supporting Evidence. In: *Machine Learning for Health (ML4H)*. 2021.
12. N. Kanwar, K. Carmine-Simmen, R. Nair, C. Wang, S. Moghadas-Jafari, H. Blaser, D. Tran-Thanh, D. Wang, **P. Wang**, et al. Amplification of a Calcium Channel Subunit CACNG4 Increases Breast Cancer Metastasis. In: *EBioMedicine*. 2020.
13. H. D. Gonorazky, S. Naumenko, A. K. Ramani, V. Nelakuditi, P. Mashouri, **P. Wang**, et al. Expanding the Boundaries of RNA Sequencing as a Diagnostic Tool for Rare Mendelian Disease. In: *The American Journal of Human Genetics*. 2019.